

Federated Learning under Structural Class Absence for Multi-City Transportation Mode Detection

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Abstract—Real-world cross-silo federated learning (FL) for transportation mode detection (TMD) reveals a challenge deeper than distributional heterogeneity: cities inhabit physically different mobility ecosystems with structurally disjoint transport mode sets. We identify this as a partially class-disjoint data (PCDD) condition and show that gradient interference from physically uninformed clients is the key failure mechanism, which can cause standard FL methods to severely degrade rare-class performance. We propose a two-level class-aware FL framework that enforces physical knowledge boundaries in both local training and global aggregation, combining class-weighted loss, knowledge inheritance, class-wise aggregation (cwFedAvg), and loss-aware client weighting (LAWA). Applied to real long-term evolution (LTE) cellular data from four Saudi cities, the method outperforms FedAvg, FedProx, SCAFFOLD, FedBN, and MOON, outperforms a standard centralized baseline under comparable settings without sharing raw measurement data, and remains reliable across dataset scales spanning an order of magnitude. This work is a problem-driven study: it identifies a structural failure mode in FL under class absence and derives principled design responses from physical reality.

Index Terms—federated learning, transportation mode detection, cellular networks, class imbalance, knowledge inheritance, heterogeneous data

I. INTRODUCTION

Transportation mode detection (TMD), identifying whether a user is walking, cycling, or driving, is fundamental for smart mobility systems. Cellular network signals offer a compelling GPS-free alternative, carrying implicit mobility information through Reference Signal Received Power (RSRP), Channel Quality Indicator (CQI), handover frequency, and timing advance [1]. Transport modes are sustained physical states: pedestrians maintain 0–2 m/s, cyclists 3–8 m/s, highway vehicles exceed 30 m/s. Our key finding is that windowed features improve macro F1 by 87% over row-by-row classification.

The multi-city nature of TMD makes FL the natural deployment framework, as cities cannot share raw trajectory data due to privacy regulations [2]. However, applying standard FL to real cross-silo Transport Mode Detection reveals a problem more fundamental than distributional heterogeneity. The four cities Jeddah, KAUST, KZ (Karaganda), and Makkah

are physically different mobility environments. KAUST is a self-contained campus where residents jog, cycle, and ride scooters (mean velocity 12.3 m/s). Jeddah is an urban city where highway driving dominates (mean velocity 32.3 m/s). Makkah has the densest cellular infrastructure (3.0 unique cells per window). These are not statistical accidents; they reflect the physical and social geography of each location.

This creates what we identify as a partially class-disjoint data (PCDD) setting. KAUST is the only city where motorcycle, jog, scooter, and bike trips occur. Standard FedAvg [3] is blind to this: clients that have never observed jog still contribute gradients that can overwrite KAUST’s learned jog representation, not because their data is wrong, but because their physical environment simply does not contain that mode. The result is predictable and experimentally confirmed: FedAvg collapses jog and motorcycle to F1 = 0.000. This cannot be fully explained by client drift alone, since the target of FedProx [4] and SCAFFOLD [5] and neither method recovers these classes. The problem requires a solution that respects what each city can physically observe.

This work identifies gradient interference from structurally uninformed clients as a key failure mechanism in cross-silo Transport Mode Detection.

The contributions of this work are:

- 1) We identify PCDD, a structurally distinct FL setting where class absence is physically determined by client environment, not sampling. We analyze failure modes of standard FL methods and establish design principles, showing that interference from uninformed clients is the dominant mechanism.
- 2) We propose a class-aware FL framework enforcing physical knowledge boundaries through knowledge inheritance, cwFedAvg, and LAWA, and validate it on real multi-city LTE data across three dataset scales with decreasing variance, confirming deployment reliability.

Existing FL methods address distributional heterogeneity (FedAvg, FedProx, SCAFFOLD), feature-space mismatch (FedBN [9]), or representation alignment (MOON [10]), all assuming clients share the same label space. Recent work on label-space heterogeneity [8] treats missing classes as a sampling artifact rather than a structural constraint imposed by the environment. Unlike prior work that treats label het-

erogeneity as a sampling artifact, we model class absence as a structural property of the data-generating process and design the learning framework accordingly.

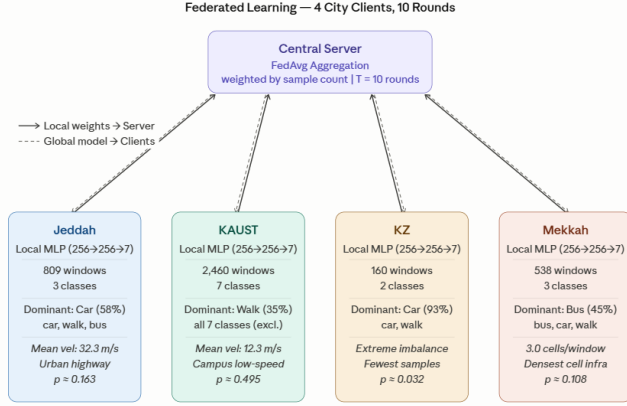


Fig. 1: Federated learning system with 4 city clients. Each client trains a local MLP on windowed cellular data and sends model weights to the central server for aggregation.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Federated Learning Setup

We consider $K = 4$ cross-silo clients (Jeddah, KAUST, KZ, Makkah). Each client k holds $\mathcal{D}_k = \{(\mathbf{x}_i, y_i)\}$ where $\mathbf{x}_i \in \mathbb{R}^{30}$ is a statistical feature vector extracted from a 60-second cellular window and y_i is the transport mode label. Clients train identical multilayer perceptrons (MLPs) [input $\rightarrow 256 \rightarrow 256 \rightarrow 7$] over $T = 10$ rounds. No raw data leaves any client.

B. Heterogeneity Challenges

Four simultaneous heterogeneity types challenge standard FedAvg: (1) Partially class-disjoint labels: KAUST exclusively holds motorcycle, jog, scooter, and bike. (2) Extreme quantity skew: KAUST has 2,460 windows vs. KZ's 160—a $15\times$ imbalance. (3) Within-client class imbalance: KZ is 93% car. (4) Physical environment heterogeneity: cities differ in velocity, handover rate, and cell density (Fig. 2).

C. Gradient Interference under Class Absence

These conditions expose a fundamental limitation of standard federated averaging under Partially Class-Disjoint Data. The global objective is:

$$F(\mathbf{w}) = \sum_{k=1}^K p_k F_k(\mathbf{w}), \quad (1)$$

where $p_k = n_k / \sum_j n_j$. Sample-proportional weighting worsens PCDD: large clients earn proportionally more influence over every class, including classes they have never observed. High data volume does not necessarily imply broad class coverage, as large clients may still observe only a subset of classes.

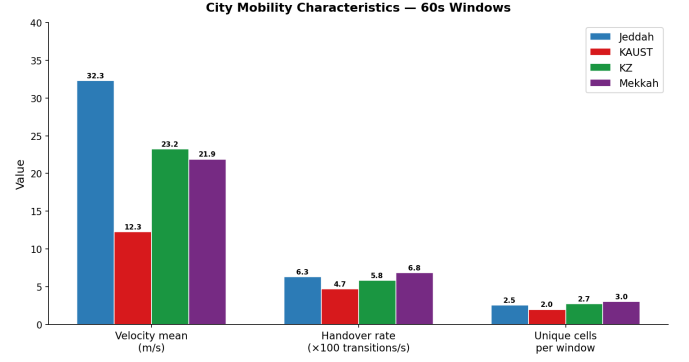


Fig. 2: City mobility characteristics. Jeddah: highest mean velocity (32.3 m/s). KAUST: lowest velocity (12.3 m/s), campus low-speed modes. Makkah: densest cell deployment (3.0 unique cells/window).

For class c , the aggregated gradient is:

$$g_c = \sum_{k=1}^K p_k g_{k,c}, \quad (2)$$

where $g_{k,c}$ denotes the gradient contribution of client k to class c . As the number of uninformed clients increases, the signal-to-interference ratio degrades proportionally. Under PCDD, clients with $n_{k,c} = 0$ receive no corrective signal for class c ; backpropagation through unrelated patterns pushes their output node actively opposed to the informed signal. This is primarily a structural data limitation rather than an optimization issue: optimization-level interventions cannot directly recover information that is absent from the local data.

The aggregated update decomposes as:

$$g_c = \underbrace{\sum_{k: n_{k,c} > 0} p_k g_{k,c}}_{\text{informed signal}} + \underbrace{\sum_{k: n_{k,c} = 0} p_k \epsilon_{k,c}}_{\text{interference}}, \quad (3)$$

where $\epsilon_{k,c}$ denotes gradient components induced by samples from other classes due to shared parameterization. For jog, KAUST contributes informed signal ($p_{\text{KAUST}} \approx 0.495$) while three cities contribute interference (≈ 0.505). The interference dominates, precisely explaining the observed collapse: $F1_{\text{jog}} = 0.000$. This decomposition highlights that interference scales with the number and weight of uninformed clients.

The framework addresses this by enforcing:

$$p_{k,c} = 0 \quad \text{if } n_{k,c} = 0, \quad (4)$$

This modifies Eq. (2) by enforcing class-wise masking of aggregation weights. Section III derives each framework component directly from this principle.

D. Feature Extraction and Physical Motivation

Each 60-second window is summarized into 30 statistical features: mean, variance, max, and min of RSRP, RSSI, RSRQ, CQI, velocity, and path loss; mean of rank indication, longitude, latitude, timing advance, PDSCH scheduled;

handover rate and unique cell count. Window label is assigned by majority vote. The 60-second scale captures one coherent mobility state, confirmed empirically: windowed features improve macro F1 by 87

III. PROPOSED METHOD

All components follow from one constraint: updates for a given class should originate only from clients that have observed that class. We formalize it via a class-availability indicator $m_{k,c} = \mathbf{1}[n_{k,c} > 0]$, where $\mathbf{1}[\cdot]$ denotes the indicator function. Locally, clients enforce $m_{k,c} = 0$ by freezing output nodes for absent classes. Globally, aggregation for class c uses only clients with $m_{k,c} = 1$. This framework operates under the assumption that clients share their class-availability indicators $m_{k,c}$ with the server. This constitutes a minor metadata leakage, the server learns which transport modes exist in each city, but no raw measurements, trajectories, or individual records leave any client. The privacy guarantee therefore applies to individual-level data rather than aggregate class structure.

A. Level 1 — Local Training

Class-Weighted Loss: Each client k assigns:

$$w_{k,c} = \frac{N_k}{C_k \cdot n_{k,c}} \quad (5)$$

where C_k is the number of classes present at client k , penalizing minority class errors proportionally. This corrects within-city physical imbalance: KZ’s geography generates 93% car trips, collapsing local training to a near-degenerate car detector without reweighting.

Knowledge Inheritance: For any class c absent at client k ($n_{k,c} = 0$), the corresponding output node weights are frozen to the global model values received at the start of that round. A city that has never produced a jog trip carries no valid signal about what jog looks like in cellular data; allowing it to update the jog node constitutes uninformed gradient interference. In practice, this is implemented by freezing the corresponding output-layer parameters during local updates. This mechanism accounts for the single largest per-component gain (+0.099 macro F1 vs. E1), recovering jog from F1 = 0.000 to 0.567. This highlights that the dominant factor is not improved optimization, but the prevention of updates from structurally uninformed clients, directly supporting the PCDD-based interpretation of the problem.

B. Level 2 — Global Aggregation

Class-Wise FedAvg (cwFedAvg): Each output node c is aggregated independently using only clients with $n_{k,c} > 0$:

$$\alpha_{k,c} = \frac{n_{k,c}}{\sum_j n_{j,c}} \quad (6)$$

We distinguish between global weights p_k , class-wise weights $p_{k,c}$, and final blended weights W_k used in aggregation. This is the server-side complement to knowledge inheritance: just as clients do not update absent-class nodes locally, the server excludes contributions from clients that have never observed

those classes globally. This ensures that each class-specific parameter is updated using only relevant data, effectively removing interference at the aggregation level.

LAWA and γ Blending: Each client computes a loss balance ratio $r_k = L_{\min,k}/L_{\max,k}$, where $r_k \rightarrow 1$ indicates balanced learning. This weighting favors clients that achieve balanced learning across their observed classes, while avoiding over-emphasis on structurally simpler clients. The final weight blends LAWA with cwFedAvg:

$$W_k = \gamma \cdot \tilde{r}_k + (1 - \gamma) \cdot \alpha_k \quad (7)$$

where $\tilde{r}_k$ are normalized LAWA weights and α_k denotes global aggregation weights and $\alpha_{k,c}$ denotes class-wise weights used at the output layer.

LAWA has a PCDD-specific limitation: KZ’s two-class simplicity yields $r_k \approx 0.92$, earning 60.9% of the LAWA signal despite holding only 3.2% of windows, while KAUST ($r_k \approx 0.065$) receives only 0.044 despite holding 62% of windows and all 7 classes. At $\gamma = 0.2$, KZ’s distortion is contained at 15.4% while Jeddah’s effective training is rewarded. Design rule: γ should be set inversely proportional to the degree of class disjointness; at full disjointness, $\gamma \rightarrow 0$ and pure cwFedAvg is optimal.

IV. EXPERIMENTAL SETUP

A. Dataset

The dataset [1] spans four Saudi cities with Long-Term Evolution devices recording at ≈ 1 Hz. Windowing yields 4,967 windows (Table I). Sliding-window variants yield 9,669 and 56,794 windows for scalability analysis (Section V-G), using strict trip-level train/test splits to prevent temporal leakage. The label space contains 7 classes: car, walk, bus, scooter, bike, motorcycle, and jog.

TABLE I: Dataset Statistics After 60-Second Windowing

City	Win.	Cls.	Dominant	Transport Modes
Jeddah	809	3	Car (58%)	car, walk, bus
KAUST	2460	7	Walk (35%)	car, walk, bus, scooter, bike, motorcycle, jog
KZ	160	2	Car (93%)	car, walk
Mekkah	538	3	Bus (45%)	bus, car, walk

B. Model, Training, and Baselines

MLP: input(30) $\rightarrow$ 256 $\rightarrow$ ReLU $\rightarrow$ 256 $\rightarrow$ ReLU $\rightarrow$ 7. SGD, lr=0.01, momentum=0.9, batch=32, 1 local epoch, 10 rounds, seed=42. Baselines: FedAvg (row-by-row and windowed), FedProx [4], SCAFFOLD [5], FedBN [9], MOON [10], FedPer, Centralized (plain CE), Centralized (class-weighted), and ablation variants.

V. RESULTS

A. Window Size Analysis

Row-by-row classification achieves F1=0.280; 60-second windows jump to 0.524 (87% improvement). Performance degrades monotonically beyond 60s (0.509 at 120s, 0.463

at 180s, 0.394 at 300s) as temporal averaging collapses the velocity and signal variability that distinguishes modes.

B. Ablation Study

All $\Delta F1$ values in Table II are cumulative improvements over E1. Class-weighted loss (E2) corrects within-city geographic imbalance (+0.033). Knowledge inheritance (E3) provides the largest per-component gain (+0.099 vs. E1; +0.066 incremental over E2), directly recovering KAUST-only classes. cwFedAvg (E4) enforces the same constraint globally (+0.092). LAWA blending (E5) achieves $F1 = 0.631$, a cumulative +0.107 over E1. These results reinforce a single mechanism: enforcing class-aware constraints that prevent gradient interference from uninformed clients, validating the PCDD hypothesis more than any individual technique.

E5 achieves mean $F1 = 0.630 \pm 0.010$ across three seeds (42, 123, 456) vs. 0.551 ± 0.021 for E1, confirming that gains are consistent and not seed-dependent. These results indicate that enforcing class-aware constraints is more critical than improving optimization dynamics alone. The lower variance of E5 indicates that class-aware constraints stabilize training.

TABLE II: Ablation Study — 60s Windows, Round 10

Experiment	Accuracy	Macro F1	$\Delta F1$ vs E1
E1: FedAvg baseline	0.7837	0.5243	—
E2: + Class-weighted loss	0.7616	0.5572	+0.033
E3: + Knowledge inheritance	0.7907	0.6234	+0.099
E4: + cwFedAvg aggregation	0.7918	0.6160	+0.092
E5: + LAWA blend ($\gamma=0.2$)	0.8048	0.6313	+0.107

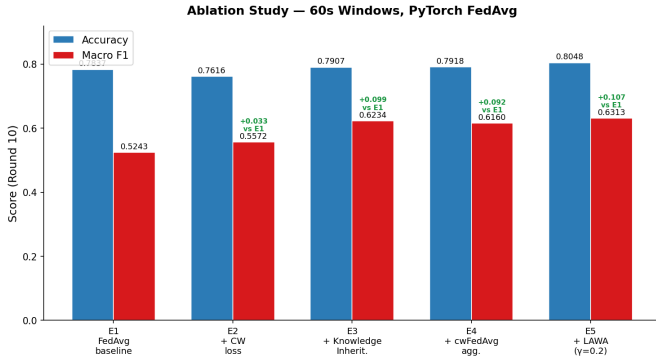


Fig. 3: Ablation study. Knowledge inheritance (E3) provides the largest single gain by preserving KAUST-only class representations that standard FedAvg destroys.

C. Per-Client and Per-Class Performance

Every city improves: KAUST most dramatically (+0.400, from 0.242 to 0.642) as the direct beneficiary of knowledge inheritance; Jeddah (+0.149), Mekkah (+0.120), and KZ (+0.122) also improve from a cleaner global feature extractor and corrected local imbalance.

Fig. 5 shows the critical PCDD result: FedAvg collapses jog to $F1 = 0.000$ (three cities contribute interference with

zero jog data), while E5 recovers it to 0.567. Motorcycle improves from 0.400 to 0.508. All other classes also improve, confirming system-wide benefit. This reflects the underlying physical characteristics: jog is a KAUST campus behavior (mean velocity 2–4 m/s, minimal handovers) that cities like Jeddah (mean 32.3 m/s) cannot observe or represent; their gradients can degrade it.

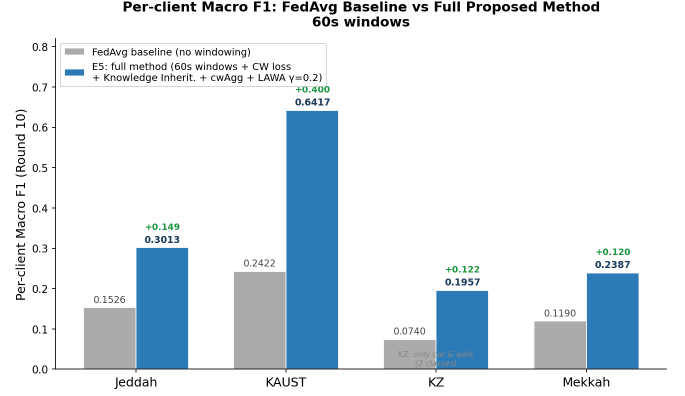


Fig. 4: Per-client macro F1: FedAvg vs. E5. All four cities improve, with KAUST benefiting most (+0.400) due to recovery of its exclusive classes motorcycle and jog.

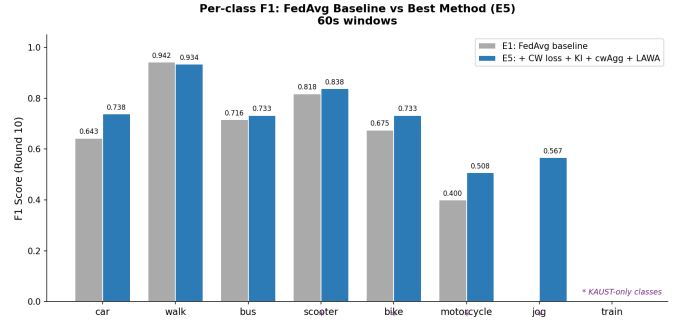


Fig. 5: Per-class F1: E1 vs. E5. FedAvg collapses jog to $F1 = 0.000$; E5 recovers it to 0.567. KAUST-only classes show the largest improvements.

D. Comparison with Baselines and Centralized Upper Bound

Table III presents the full comparison. FedBN and MOON were evaluated over 3 seeds; F1 shown for seed = 42; $\pm$ Std across seeds.

FedProx ($F1 = 0.522$) is virtually identical to FedAvg, with only 1 local epoch, client drift is negligible. SCAFFOLD ($F1 = 0.366$) performs worse than FedAvg. While SCAFFOLD has not fully converged at round 10, its oscillating trajectory, alternating between improvement and degradation across rounds, indicates that its poor performance stems from structural incompatibility with momentum-based optimization rather than insufficient training. This is consistent with prior work showing SCAFFOLD requires careful hyperparameter tuning beyond the scope of this comparison.

FedBN improves feature distribution alignment but still aggregates output nodes from clients that have never observed certain classes, leaving rare classes such as jog weakly modeled and exhibiting higher variance ($\sigma = 0.035$ vs. 0.010 for E5). FedBN achieves jog F1=0.333 (seed 42), which is barely above FedAvg’s complete collapse—while E5 recovers jog to 0.567. This failure is structurally expected under PCDD: local batch normalization addresses feature distribution shift but leaves output nodes aggregated uniformly, meaning Jeddah, KZ, and Mekkah still corrupt the jog representation. The multi-seed evaluation confirms this structural fragility: FedBN’s variance is $3.5\times$ higher (0.035 vs. 0.010), the direct signature of a method that does not directly address the class-absence issue under PCDD conditions.

MOON (F1=0.560) moderately helps representation quality but does not address the root cause: uninformed clients still update output nodes for classes they have never observed. FedPer achieves high accuracy (0.8121) but its macro F1 (0.5845) remains below E5: while FedPer gives KAUST a personalized head, the shared feature extractor is still aggregated with standard FedAvg, leaving it exposed to gradient interference from uninformed cities.

These results indicate that the primary limitation of existing methods is not optimization instability or representation misalignment, but the absence of structural class awareness under PCDD conditions, a failure mode that standard FL methods are not designed to address.

To provide a fair centralized comparison, we evaluate two models: plain CE (F1=0.591) and class-weighted centralized (F1=0.653). E5 (F1=0.631) outperforms plain CE by +0.040 and falls short of the class-weighted model by only 0.022, without sharing raw measurement data.

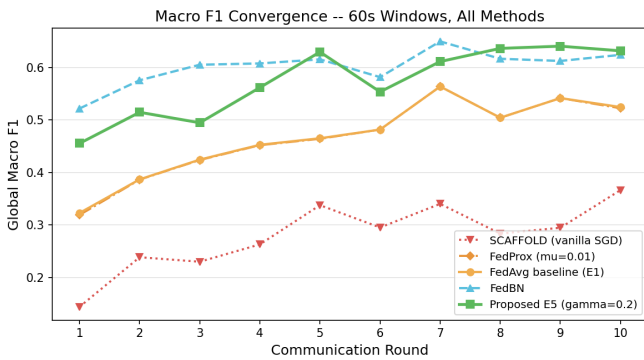


Fig. 6: Macro F1 convergence over 10 communication rounds. SCAFFOLD exhibits persistent instability throughout, consistent with the known incompatibility of its variance-reduction mechanism with momentum-based SGD. The oscillating trajectory indicates structural instability rather than insufficient training rounds. FedAvg, FedProx, FedBN, and E5 all plateau by round 7.

E. Effect of Class Disjointness (PCDD Analysis)

We construct three controlled disjointness settings by modifying class availability across clients, measuring disjointness by average clients per class. Condition A (Low, avg 2.71 clients/class): rare classes shared across clients. Condition B (Medium, avg 2.14): natural dataset configuration. Condition C (High, avg 1.86): rare classes confined to one client.

E5 consistently outperforms FedAvg across all conditions (Fig. 7). The largest gain occurs under moderate disjointness (Condition B, $\Delta = +0.114$), consistent with the theoretical PCDD expectation: when classes are present but confined to a single client, interference is maximized and class-aware aggregation is most critical. The method’s advantage peaks precisely in the regime it was designed for, validating the PCDD framing directly. This experiment isolates the effect of class availability from other forms of heterogeneity.

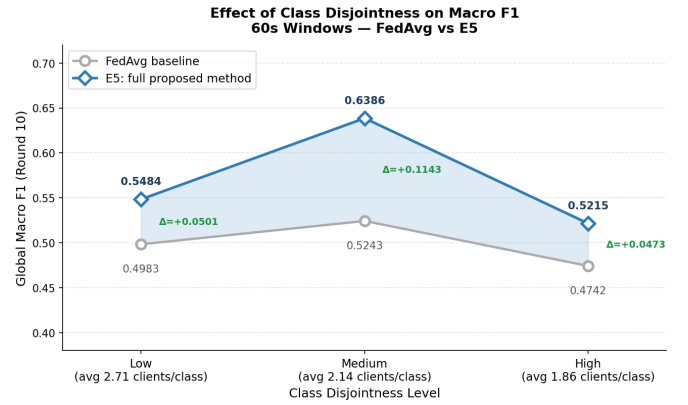


Fig. 7: Macro F1 vs. class disjointness. Peak gain (+0.114) at moderate disjointness validates the PCDD framing—the method’s advantage peaks precisely in the regime it was designed for.

F. Effect of the Blend Parameter γ

We swept $\gamma \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$ over 3 seeds (Fig. 8). $\gamma = 0.2$ is consistently the best-performing value. Performance declines monotonically for $\gamma \geq 0.3$: excessive LAWA weight allows KZ’s artificially high convergence ratio to distort the global feature space, confirming the design rule

TABLE III: Full Method Comparison — 60s Windows, Round 10. † Evaluated over 3 seeds (42, 123, 456); F1 shown for seed=42; $\pm$ Std across seeds.

Method	Accuracy	Macro F1	$\pm$ Std	vs FedAvg
SCAFFOLD	0.7163	0.3663	—	−0.158
FedProx	0.7827	0.5220	—	−0.002
FedAvg baseline	0.7837	0.5243	—	—
MOON†	0.8008	0.5600	0.010	+0.036
FedPer	0.8121	0.5845	—	+0.062
Centralized-CE	0.8421	0.5908	—	+0.067
FedBN†	0.8038	0.6237	0.035	+0.099
Proposed E5†	0.8048	0.6313	0.010	†+0.107
Centralized-CW	0.8260	0.6527	—	+0.128

that γ should decrease with class disjointness. While $\gamma = 0.2$ is identified empirically, the design rule that γ should decrease with class disjointness suggests a natural estimator: γ could be set proportional to the average class overlap across clients, computed from the shared mk,c indicators.

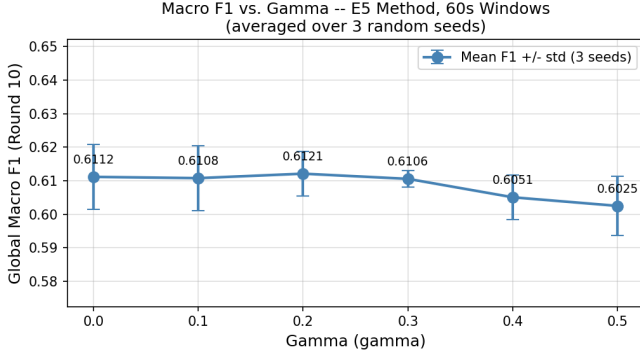


Fig. 8: Macro F1 vs. γ (3 seeds). $\gamma = 0.2$ achieves the highest mean F1 and lowest variance. Performance degrades monotonically for $\gamma \geq 0.3$.

G. Data Scalability Analysis

We evaluate E5 vs. E1 across three scales with strict trip-level splits (Table IV, Fig. 9). Note that lower absolute F1 at sliding-window scales reflects the stricter trip-level protocol, not performance degradation. E5 outperforms FedAvg at every scale, improves as data grows ($0.5551 \rightarrow 0.5751$), and its variance decreases monotonically ($\sigma: 0.010 \rightarrow 0.012 \rightarrow 0.009$) while FedAvg’s remains higher ($\sigma = 0.017$ at maximum scale). E5’s structural constraints provide the deployment consistency that random initialization cannot.

TABLE IV: E5 vs. FedAvg Across Dataset Scales (Trip-Level Split, 3 Seeds).

Dataset	Win.	FedAvg	E1 σ	Proposed F1	E5 σ
Non-overlapping	4,967	0.5509	0.021	0.6301	0.010
Sliding 30s-step	9,669	0.5192	0.021	0.5551	0.012
Sliding 5s-step	56,794	0.5600	0.017	0.5751	0.009

Limitations. The current study involves four clients reflecting a realistic but limited cross-silo scenario. Behavior under partial class overlap and scalability to larger client populations remain open directions. The framework applies a binary class-availability criterion; behavior under very sparse class observations (e.g., one or two samples) warrants further study, as gradients from such clients may be unreliable despite being non-zero.

VI. CONCLUSION

We identified PCDD, a structural FL failure mode where class absence is physically determined by client environment, and showed that gradient interference from uninformed clients can severely degrade rare-class performance. Our class-aware framework enforces physical knowledge boundaries at

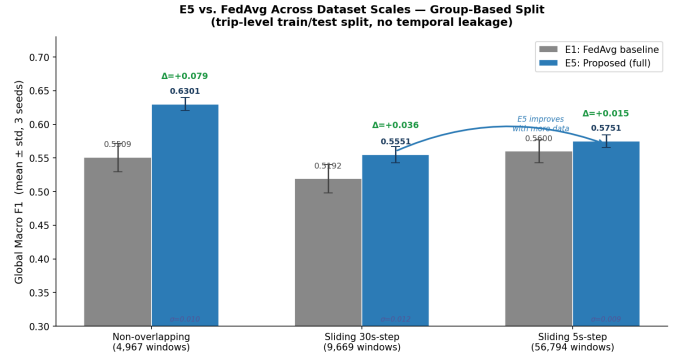


Fig. 9: E5 vs. FedAvg across three scales (trip-level split, mean $\pm$ std over 3 seeds). E5 consistently outperforms FedAvg, improves with more data, and variance shrinks monotonically.

both local and global levels, outperforming all standard FL baselines and approaching a class-aware centralized upper bound (within 0.022 macro F1) without sharing raw measurement data across city boundaries. The multi-seed evaluation ($F1 = 0.630 \pm 0.010$ for E5 vs. 0.551 ± 0.021 for FedAvg) confirms consistent gains. The advantage holds across dataset scales spanning more than an order of magnitude, with E5 variance decreasing monotonically, confirming suitability for real-world deployments where cellular data accumulates continuously. These results suggest that enforcing class-aware constraints is essential for robust FL under structural class absence. Future work will explore larger and more dynamic client populations.

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